

Integrated Planar Transformer for High-Voltage-Ratio LLC DCX With High Current and Ultra-Low Voltage Output

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ABSTRACT The 48 V voltage regulation modules (VRM) for microprocessors feature a high voltage transformation ratio, large current, and ultra-low voltage output. The solutions based on high-turns-ratio transformers are the most popular ones. However, these transformers often suffer from significant winding loss and large size issues, which are the main bottlenecks in improving efficiency and power density. This paper proposed a highly integrated planar transformer suitable for high-voltage-ratio VRMs with high current and ultra-low voltage output. The transformer integrates with the SRs and output capacitors and has a 1/10 turn on the secondary side, which can effectively reduce winding loss and improve power density. The winding length is determined by the size of the SRs and the output capacitors, resulting in a quite different design and optimization process from traditional transformers. The constraints of the transformer structure are analyzed and a flexible and compact structure is achieved on a 6-layer PCB. Based on the proposed transformer, a 40 V–1 V/200 A, 1 MHz LLC DCX is built. Experimental results show that the LLC DCX achieves a peak efficiency of 97% and a power density of 1394 W/in³, surpassing the state-of-the-art transformer-based solutions.

INDEX TERMS LLC DCX, magnetic integration, planar transformer, voltage regulation module (VRM).

I. INTRODUCTION

The 48 V voltage regulation module (VRM) for data centers is a current research hotspot. It converts the bus voltage of 40 V–60 V into ~1 V to power the microprocessor. Such a high input-output voltage ratio makes it very challenging to improve the efficiency and power density of VRM [1], [2]. The solutions based on high-turns-ratio transformers are the most popular ones. Fig. 1 shows two popular 48 V VRM solutions, where the DC transformer (DCX) is an unregulated converter based on a high-turns-ratio transformer. The high-turns-ratio transformer usually accounts for the largest size and loss in the DCX, which are the main bottlenecks in improving efficiency and power density.

In addition to the high voltage ratio, the DCX has an ultra-low output voltage (~1 V) and a quite large output current

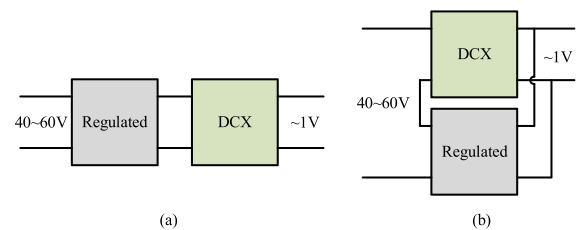


FIGURE 1. Popular 48 V VRM solutions, where the DCX is based on a high-turns-ratio transformer. (a) Two-stage solution. (b) Single-stage sigma solution.

(~100 s of Amps). At high frequency and low output voltage, the core losses are small due to the small volt-second product. Meanwhile, the primary winding requires many turns to

achieve a high voltage ratio, and the secondary side requires many synchronous rectifiers (SRs) in parallel to handle large currents, which results in large winding losses and termination losses. Therefore, the winding losses and termination losses often account for the largest proportion in such applications [3], [4].

Matrix transformers are proposed for the DCXs, which are suitable for high-frequency and large current applications [4], [5], [6]. The matrix transformer uses multiple secondary windings wrapped around multiple magnetic legs and the SRs are distributed on the windings. A decentralized layout structure reduces current stress and current sharing problems in windings so that winding losses and termination losses can be reduced [7]. However, the matrix transformer is an integer-turn winding structure, and its winding losses are still significant in the DCXs. Reference [4] designed a high-density 40 V–1 V/80 A LLC DCX with a four-legs matrix transformer. Reference [5] proposed a 12-leg planar matrix transformer for 48 V–1 V/300 A LLC DCX. Both studies are based on planar matrix transformers, and the winding losses are much higher than the core losses. The peak efficiency is only around 94% in both studies.

Fractional turn transformers can further reduce winding losses by implementing winding structures of less than 1 turn. The turns of secondary winding N_s in the LLC circuit are calculated as follows:

$$N_s = \frac{V_o T_s}{4A_e B_m} \quad (1)$$

where V_o is the output voltage, T_s is the switching period, A_e is the effective cross-sectional area of the magnetic core, B_m is the peak flux density. As the switching frequency increases and the output voltage decreases, the volt-second product on the transformer winding becomes smaller, and thus the number of winding turns decreases accordingly. Fractional-turn transformers are quite suitable for high-frequency and extremely low-voltage situations. In recent years, with the application of wide bandgap devices and the continuous improvement of switching frequency, fractional turn transformers have received a lot of attention. Current research mainly focuses on planar fractional-turn transformers in 380 V–12 V applications with 1/2 and 1/4 turn symmetry structures [8], [9], [10], [11], [12] and verifies the advantages of fractional turns in improving efficiency and power density. Reference [13] mainly focuses on the modeling of asymmetric fractional-turn transformers and are verified using a 1/3-turn transformer. However, the planar fractional-turn transformer has not yet been reported in 48 V VRM applications with ultra-low voltage output and smaller fractional turns.

This paper proposed an integrated planar transformer for high-voltage-ratio LLC DCX with high current and ultra-low voltage output. The proposed planar transformer achieves a 1/10 turn on the secondary side to reduce the winding loss. The compact integration with SRs and output capacitors can

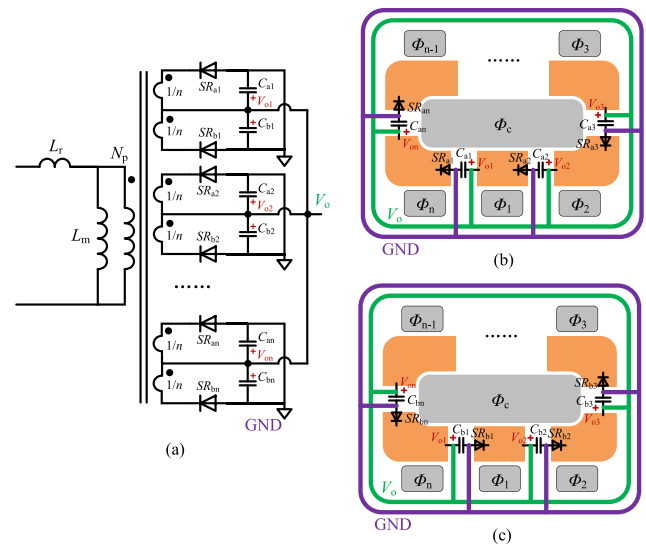
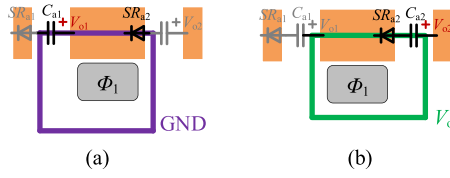
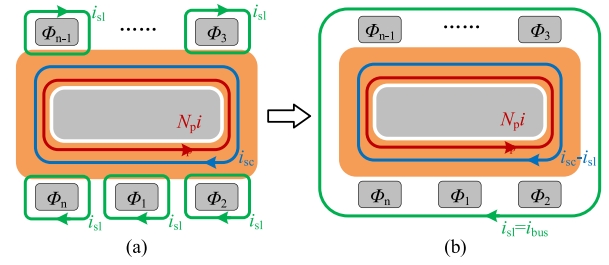
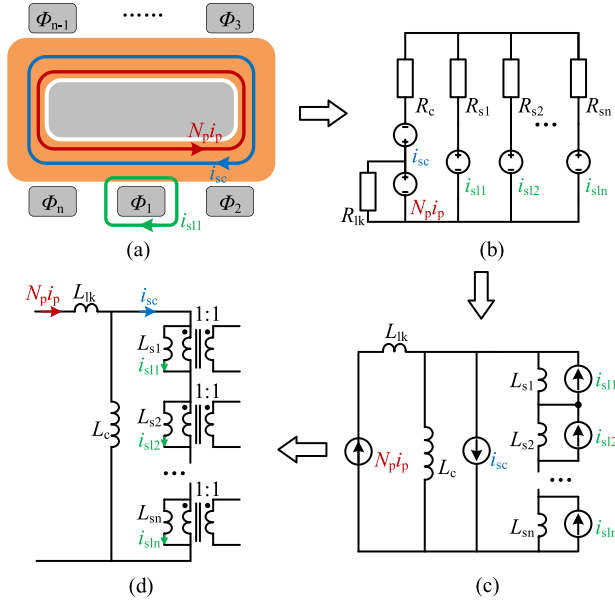


FIGURE 2. Schematic layout of the proposed integrated transformer. (a) Integrated transformer circuit. (b) Top layer. (c) Bottom layer.

further reduce winding losses and increase power density. The winding length is determined by the size of the SRs and the output capacitors with ultra-low voltage output, resulting in a quite different design and optimization process from traditional transformers. The constraints of the transformer structure are analyzed and a flexible and compact structure is achieved. Besides, the proposed transformer is implemented on a 6-layer PCB board, which reduces cost and design complexity compared to other solutions with more PCB layers. Based on the proposed transformer, a 40 V–1 V/200 A, 1 MHz LLC converter is built. Experimental results show that the LLC converter works well, achieving 97% peak efficiency and 1394 W/in³ power density, which are better than the state-of-the-art transformer-based solutions.

II. ANALYSIS OF INTEGRATED PLANAR TRANSFORMER

Fig. 2 shows the circuit and schematic layout of the proposed fractional-turn transformer with integrated SRs and output capacitors. The magnetic core has n side legs and 1 center leg. There are n groups of secondary rectifiers and output capacitors distributed on the secondary windings, as shown in Fig. 2(b) and (c). Diode symbols are used for SRs in the figures for drawing simplicity. The output V_o and GND buses are closed loops and compactly laid out around the magnetic core. All output capacitors are connected to the output V_o and GND buses. The output capacitors of each group are assumed to be large enough to provide a low-impedance path for the secondary winding current. The top SRs are all turned on in the positive half cycle, while the bottom SRs are all turned on in the negative half cycle. The secondary winding current flows the shortest path around the center leg. For each side leg, two current loops can be found, as shown in Fig. 3. According


FIGURE 3. Flux in each side leg is clamped by output voltages.

FIGURE 5. Simplified current loops.

FIGURE 4. Modelling process. (a) Current loops. (b) Magnetic circuit. (c) Electric circuit. (d) Equivalent circuit.

to Faraday's law, it can be deduced:

$$\begin{cases} \frac{d\Phi_1}{dt} = V_{o1} = V_{o2} \\ \frac{d\Phi_2}{dt} = V_{o2} = V_{o3} \\ \dots \\ \frac{d\Phi_n}{dt} = V_{on} = V_{o1} \end{cases} \Rightarrow \begin{cases} V_{o1} = V_{o2} = \dots = V_{on} \\ \Phi_1 = \Phi_2 = \dots = \Phi_n \end{cases} \quad (2)$$

It can be concluded that when all output capacitors are connected to output V_o and GND buses outside the transformer, all output capacitor voltages are naturally clamped the same and the magnetic flux in the side leg is naturally the same, which does not depend on whether the transformer structure is symmetrical or not.

For the center leg, we can deduce that:

$$\frac{d\Phi_c}{dt} = \sum_{k=1}^n V_{ok} = nV_o \Rightarrow V_o = \frac{1}{n} \frac{d\Phi_c}{dt} \quad (3)$$

The output voltage is $1/n$ of the induced voltage around the center leg, so the equivalent number of turns on the secondary side is $1/n$. The flux in each side leg is $1/n$ of the flux in the center leg. The transformer turns ratio is $N_p:1/n=N_p*n:1$.

Fig. 4(a) shows the current loops in the proposed fractional-turn transformer. There are two current loops around the center leg: the primary current $N_p i_p$ and the secondary current i_{sc} , both of which have the shortest loop path. i_{sln} is the current that flows through the loop around the side legs.

Fig. 4(b) is the magnetic circuit. R_{sn} is the reluctance of the side legs. Each side-leg should be kept the same for symmetry. R_c is the reluctance of the central leg. Then the magnetic circuit model is transformed into an electric circuit model through topological duality [14], [15], as shown in Fig. 4(c). The magnetomotive forces are transformed into current sources. The inductances are the reciprocal of reluctance. Further, the electric circuit can be converted into an equivalent circuit as shown in Fig. 4(d). The current sources are replaced by ideal transformers to achieve physical connections. From the equivalent circuit, we can see that this is a structure with the primary side connected in series and the secondary side connected in parallel. When the side-leg inductance $L_{s1}=L_{s2}=\dots=L_{sn}=L_s$ (that is $R_{s1}=R_{s2}=\dots=R_{sn}$), the current of each output is naturally shared.

The magnetizing inductance L_m is a combination of the side-leg inductance L_s and the center-leg inductance L_c .

$$L_m = N_p^2 (L_c \parallel L_s) \quad (4)$$

Fig. 5 shows the simplified current loops. When the magnetic resistance of all side legs is equal, the current between each two side legs is equal in magnitude and opposite in sign and cancels each other. It can be further simplified to a large bus loop outside the magnetic core. The current i_{sl} is equal to i_{bus} . The circulation current on the output buses i_{bus} can be regarded as part of the magnetizing current, which is equal to the current flowing through the side-leg inductor:

$$i_{bus} = i_{sl} = \frac{1}{L_s} \int n v_o dt \quad (5)$$

Because the GND and V_o buses are shorted loops, i_{bus} is loaded with the parasitic resistance R_{bus} of the GND and V_o buses. The loss caused by the circulation current is $i_{bus}^2 * R_{bus}$. In LLC DCX, the magnetizing current is usually much smaller than the winding current. Therefore, the circulation current is relatively small and has little impact on the loss.

TABLE 1. Comparison of Candidate SR Devices

	GaN	Si		
Types	EPC2066	SiR178DP	PI5101-01-LGIZ	IQE004NE1LM7
V_{DS} (V)	40	20	5	15
$R_{ds(on)}$ @125°C (mΩ)	1.32	0.47	0.45	0.61
Q_g @5V (nC)	21.8	82	56	24
Q_{oss} @4V (nC)	6	17	13.85	10
Footprint (mm ²)	13.9	31.67	32.8	10.89
$R_{ds(on)} * Q_g$ (mΩ*nC)	28.8	38.6	25	14.66
$R_{ds(on)} * Q_{oss}$ (mΩ*nC)	7.9	8.01	6.18	6.11

III. DESIGN AND OPTIMIZATIONS

To verify the proposed fractional-turn transformer, an unregulated LLC converter with a 40:1 voltage ratio is designed. The output is 1 V/200 A. The primary uses a half-bridge and the secondary uses full-wave rectification. The transformer has 2 turns of the primary winding and 1/10 turn of the secondary winding. There are 10 groups of outputs, and the output current of each output is 20 A. So the total output current is 200 A. The design procedure of LLC DCX can refer to [16]. Based on this, the parameters of the transformer are calculated. The magnetizing inductance L_m is designed to achieve ZVS of the power switches under a given dead time. L_m gets the required value by adjusting the air gap of the magnetic core. The leakage inductance of the transformer is used as the resonant inductance L_r . Since it works in DCX mode without voltage regulation capability, there is no constraint on the resonant inductor. Considering the high dynamic requirements of VRM, L_r should be as small as possible.

The drain-source voltage stress on the SRs is only 2 V, so the SRs need to be selected with ultra-low breakdown voltage. Besides, the on-resistance of the SRs needs to be small enough to reduce the conduction loss. Table 1 compare some candidate SR devices. The Si device IQE004NE1LM7 has the best overall performance with the smallest packaging area and figure-of-merit ($R_{ds(on)} * Q_{oss}$), so it is selected as the SRs. GaN devices with low on-resistance only have a breakdown voltage of 30–40 V on the market, and their FOM is larger than the best Si devices in this application. Besides, the “body diode” voltage drop of GaN devices is larger than that of Si MOSFETs, causing larger deadtime loss. Therefore, we choose Si devices for the SRs.

Fig. 6 shows the loss comparison of magnetic materials at 1 MHz and 100 °C. The loss curves are extracted from the manufacturer’s magnetic materials datasheet. Among them, DMR53 from DMEGC Company has the smallest magnetic loss per unit volume at 1 MHz and was selected in the design.

A core with a circular center leg has the shortest winding length for a given cross-section area A_e . We start the design with a magnetic core that has a circular center leg, as shown in Fig. 7.

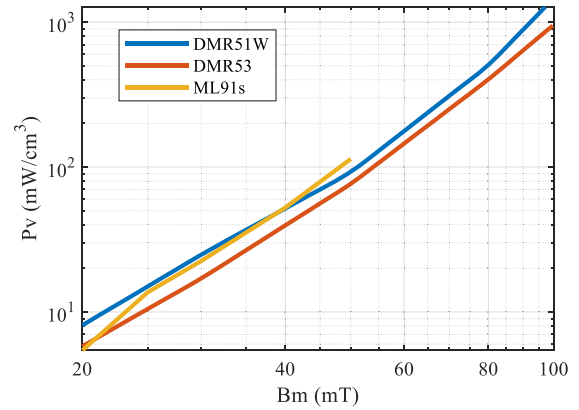


FIGURE 6. Loss comparison of magnetic materials at 1 MHz and 100 °C.

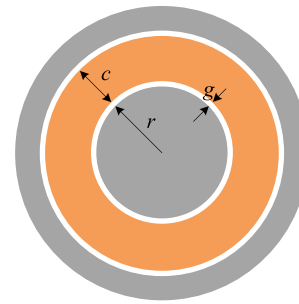


FIGURE 7. Magnetic core with a circular center leg.

The induced voltage on a winding around the center leg is nV_o when the SRs are on. According to Faraday’s law, the relationship between the effective cross-sectional area A_e of the center leg and the magnetic flux density B_m can be deduced in the LLC circuit:

$$A_e(n) = \frac{nV_o T_s}{4B_m} \quad (6)$$

Then we can get the radius r of the central leg:

$$r(n) = \sqrt{\frac{A_e}{\pi}} = \sqrt{\frac{nV_o T_s}{4\pi B_m}} \quad (7)$$

The average length of secondary winding l_s for each output is

$$l_s(n) = \frac{2\pi(r + g + \frac{c}{2})}{n} = \sqrt{\frac{\pi V_o T_s}{n B_m}} + \frac{\pi(2g + c)}{n} \quad (8)$$

where c is the winding width, g is the clearance between the windings and cores.

The length of the primary winding l_p is:

$$l_p = N_p l_s(n) \quad (9)$$

where N_p is the turns of the primary winding. It can be seen that the winding lengths of both the primary side and the secondary side are reduced with the increase of n .

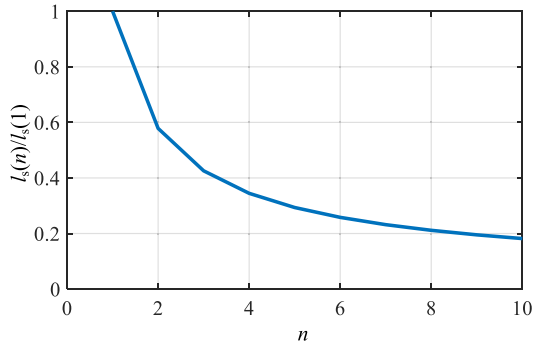


FIGURE 8. Relationship of winding length with n .

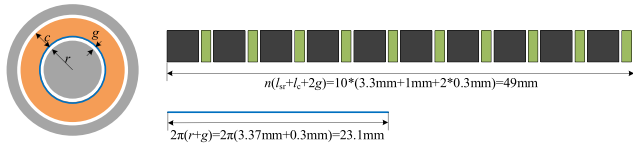


FIGURE 9. The Winding length is too short to put the SRs and output capacitors on it.

TABLE 2. Footprint of the Three Schemes

Scheme	A	B	C
Footprint area (mm ²)	394.2	311.4	398.5

Fig. 8 shows the relationship of winding length with n . The winding length in the case of $n=10$ is only 18% of that in the case of one integral turn, resulting in significant winding loss reduction. The matrix transformer is a type of integer-turn transformer, which is the case of $n=1$.

The footprint of SR is 3.3*3.3 mm. The output capacitors are 0402 packages with a footprint of 1 mm*0.5 mm. The clearance between the output capacitors and SRs is set at 0.3 mm, as well as the clearance between the winding and the magnetic core. The length of 10 sets of SRs and capacitors is 49 mm. The inner circumference of the winding is only 23.1 mm when $B_m=70$ mT, which is too short to place the SRs and output capacitors on it, as shown in Fig. 9.

The minimum length of the winding is determined by the package size of the SRs and the output capacitors. To put all SRs and output capacitors on windings and to reduce the transformer footprint, we redesigned the center leg to a rectangular shape. There are some considerations when laying out: 1) place a side leg core near each SR and lay out the GND and V_o bus compactly; 2) Avoid placing side leg cores outside the output capacitors to lead out to the output buses.

Fig. 10 shows three candidate schemes. In scheme A, all SRs and output capacitors are placed on two sides of the center leg core. The winding length and board area of this scheme are relatively large. In Schemes B and C, the SRs and output capacitors are placed on the four sides of the center leg core, and their winding length is the shortest. Scheme C has a larger center leg area and lower corner utilization than Scheme B, so it occupies a larger board area. Table 2 compares the

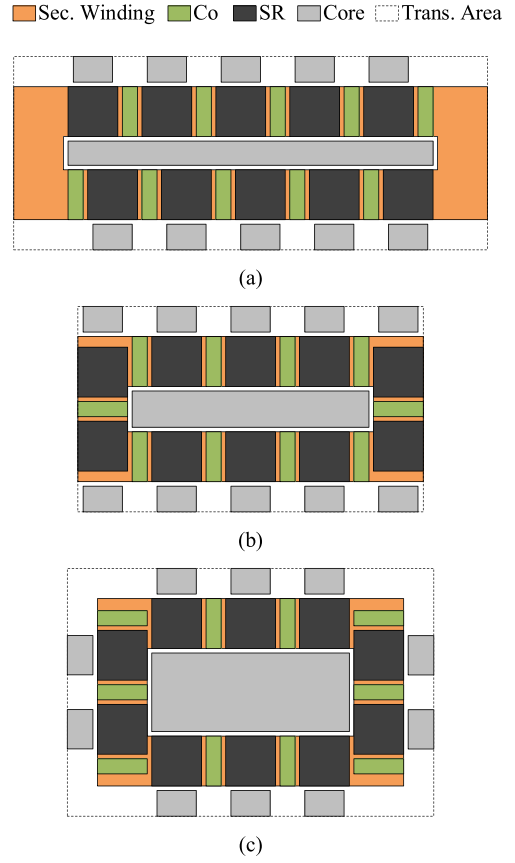


FIGURE 10. Three candidate schemes. (a) Scheme A. (b) Scheme B. (c) Scheme C.

TABLE 3. Circuit Parameter

Parameters	Values	Parameters	Values
V_{in}	40V	T_d	35ns
V_o	1V	L_m	2.2μH
I_o	200A	C_r	600nF
f_s	1MHz	L_r	35nH
C_o	2040μF@1V	Sec. driver	2EDN7524GXTMA1
Pri. switch	EPC23103 (100V)	Sec. switch	IQE004NE1LM7 (15V)

footprint area of the three schemes. Scheme B has the smallest footprint, so it is chosen.

Fig. 11 shows the model for Scheme B. The parameters b , g , d , h_1 are fixed values, which are determined by the devices' packaging size and layout. In this paper, $b=15.7$ mm, $g=0.3$ mm, $d=2.6$ mm, $h_1=2.9$ mm. The parameters a and c are adjustable variables. c is the winding width. a is the width of the center-leg core, which determines the cross-sectional area of the center-leg core because $A_e=a*b$. The remaining parameters e and h_2 can be calculated based on the existing parameters. We set the sum of the cross-sectional areas of all

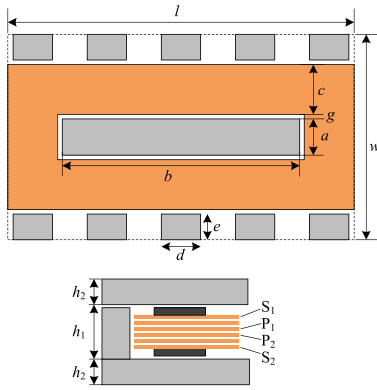


FIGURE 11. Model for Scheme B.

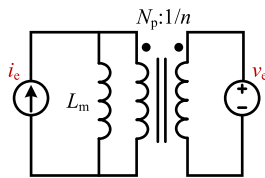


FIGURE 12. Simulation model in Maxwell.

side legs to be equal to A_e , and then we can get:

$$e = \frac{ab}{10d} \quad (10)$$

The thickness of the top and bottom plates are calculated based on the cross-sectional area:

$$h_2 = \frac{ab}{b + g + c} \quad (11)$$

The total footprint of the transformer is

$$\begin{aligned} A_T &= wl = (b + 2(g + c))(a + 2(2g + c + e)) \\ &= (b + 2(g + c)) \left(a + 2 \left(2g + c + \frac{ab}{nd} \right) \right) \end{aligned} \quad (12)$$

By sweeping parameters a and c , the winding loss, core loss, and transformer footprint curves can be plotted. The transformer footprint is calculated through the above formula. The winding loss and core loss are extracted through Maxwell 3D simulation with an eddy current field. To extract the winding loss and core loss simultaneously, the excitation in Maxwell is set according to Fig. 12. The primary side is set up with a current source excitation, and the secondary side is set up with a voltage source excitation. The secondary current can be naturally induced. The setting of the current source excitation on the primary side determines the winding losses. It should be noted that the peak value of the excitation source is set in Maxwell. According to the Steinmetz equation, the core loss per unit volume is determined by the peak magnetic flux density B_m at a fixed frequency. The excitation voltage

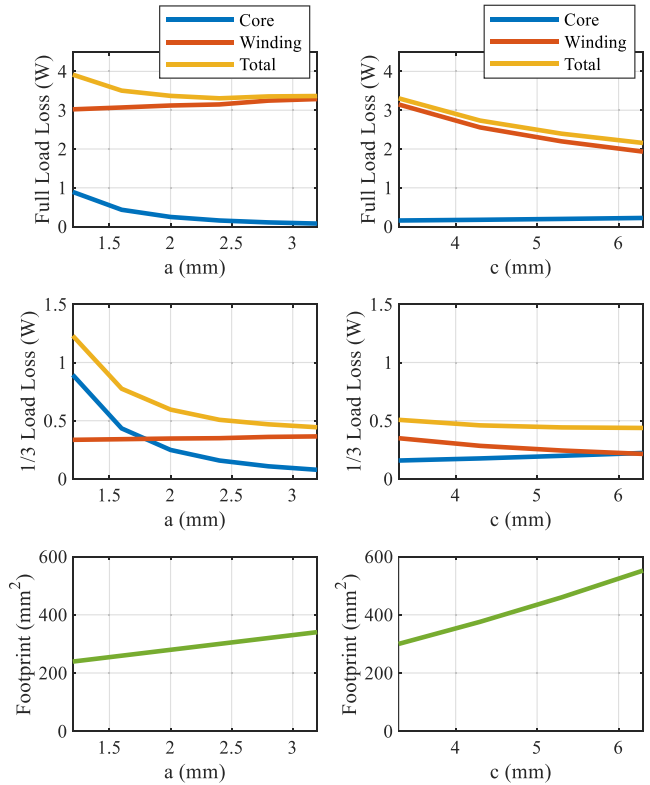


FIGURE 13. Transformer optimization.

v_e is set on the secondary side and the voltage value is determined by the peak magnetic flux density B_m . Assuming $B(t) = B_m \sin(\omega t)$, we can get

$$v_e = \frac{d(A_e B_m \sin \omega t)}{dt} = \omega A_e B_m \cos \omega t \quad (13)$$

where A_e is the effective area of the magnetic core. The peak value of excitation voltage can be calculated by $v_{em} = \omega B_m A_e$. Combining (6), it can be further derived:

$$v_e = \frac{nV_o \pi}{2} \cos \omega t \quad (14)$$

Fig. 13 shows the curves of the transformer's loss and footprint vary with the core width a and winding width c . As a increases, A_e increases linearly. On the one hand, the core loss is reduced because of the reduction of the peak flux density B_m . On the other hand, the footprint and winding losses of the transformer increase accordingly. The core loss and footprint are increased with the increase of c , while the winding loss decreases accordingly.

The core loss is a fixed loss and does not change with the load current, while the winding loss decreases with the increase of the load current. The transformer footprint is weighed against the total loss at 1/3 load. Finally, $a = 2.4$ mm and $c = 3.3$ mm are chosen.

A 6-layer PCB board is used, where the top and bottom layers are used to place the secondary windings, the 3rd and

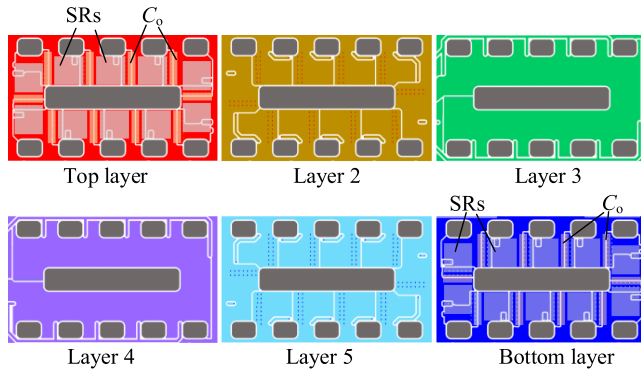


FIGURE 14. PCB layout.

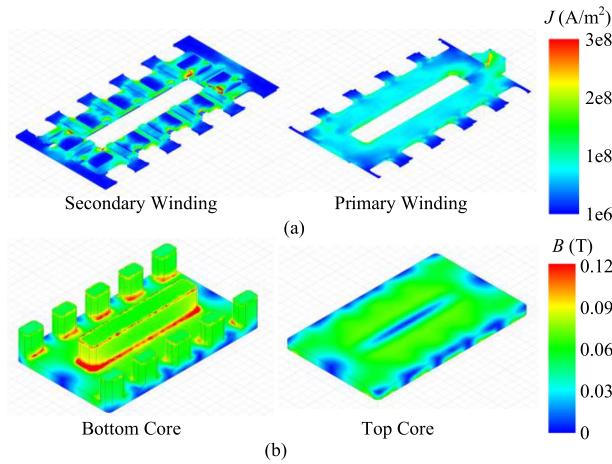


FIGURE 15. Simulation results. (a) Current density in windings. (b) Flux density in magnetic cores.

4th layers are used to place the primary windings, and the 2nd and 5th layers are used for interconnections. The copper thickness of all layers is 2oz. The PCB layout is shown in Fig. 14.

The transformer is simulated in 3D Maxwell. Fig. 15 shows the simulated current density in windings and flux density in magnetic cores, both of which are well-symmetrical and are as expected. Besides, the secondary winding is full of SRs and C_o . The leads of SRs and C_o have thick copper and are soldered on the secondary windings, which can effectively enhance the current capacity of the windings and further reduce winding losses.

IV. EXPERIMENTAL VERIFICATIONS

A 40 V-1 V/200 A LLC DCX is built to verify the proposed integrated fractional-turn transformer. The experimental prototype is shown in Fig. 16. The primary uses a half-bridge GaN chip with integrated drivers. The SRs are 15 V Si devices with a packaging size of 3.3 mm*3.3 mm. The total output capacitance is 2040 μ F at 1 V output voltage. The SRs and C_o fills the seconding winding. Two secondary drivers are connected directly in parallel to enhance the driving capability.

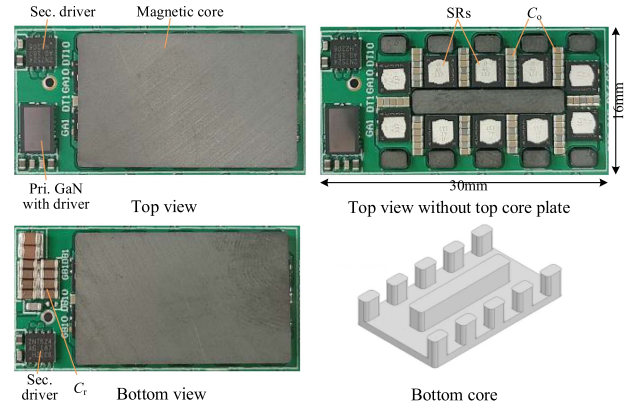


FIGURE 16. Experimental prototype.

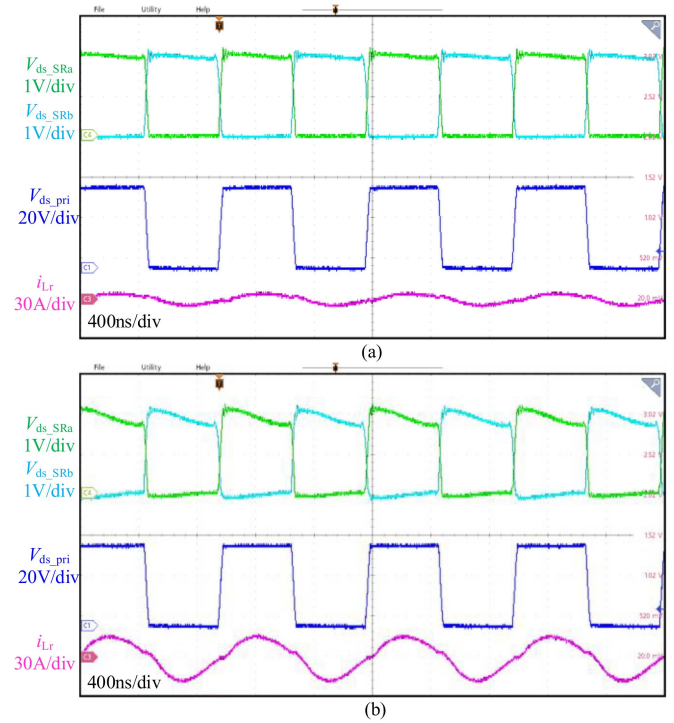


FIGURE 17. Measured key waveforms. (a) 50 A load. (b) 200 A load.

The power density of the LLC converter is 1394 W/in³ with a size of 16 mm *30 mm* 4.9 mm.

Fig. 17 shows the measured key waveforms at 25% load and full load. The LLC converter works well and the measured waveforms are very clean. The control timing is carefully tuned so that all switches achieve ZVS with almost no body diode conduction. Fig. 18 shows the measured waveforms of SRs. The v_{gs} and v_{ds} waveforms of different SRs overlap well, indicating that the proposed integrated transformer works well.

Fig. 19 shows the measured efficiency. The peak efficiency is 97% achieved at 40 A output current. The full load efficiency is 91.8%. The efficiency remains above 94% in the

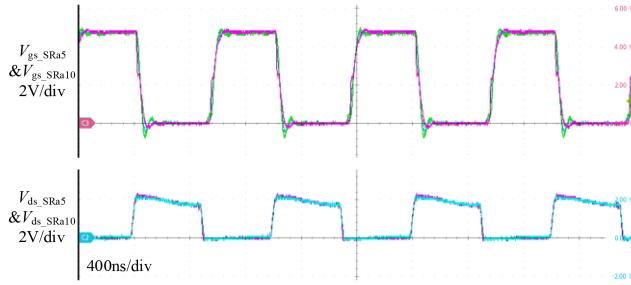


FIGURE 18. Measured waveforms of SRs.

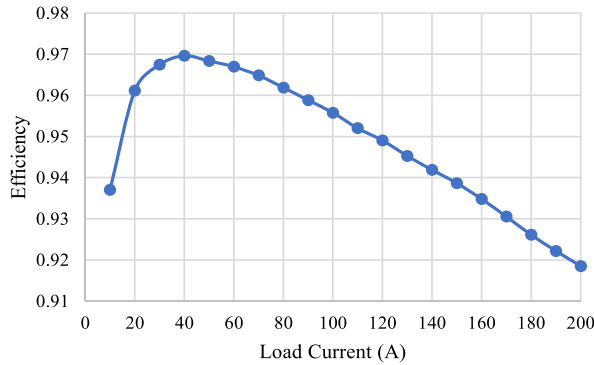


FIGURE 19. Measured efficiency.

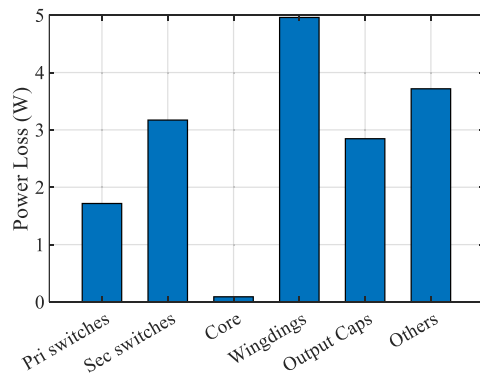


FIGURE 20. Loss breakdown under full load.

range of 10%-70% load. The efficiency curve doesn't include the driving loss.

Fig. 20 shows the loss breakdown under full load. The loss breakdown takes into account the influence of temperature. The winding losses and switch conduction losses dominate under full load. The estimated loss is lower than the measured total loss, and the missing part is classified as others. The others may include losses caused by welding and connection, etc. At higher load current, I^2R losses account for a larger proportion and become more dominant, causing efficiency to decrease. Selecting power switches with smaller on-resistance or widening the winding width can reduce I^2R losses. In addition, the on-resistance of the power switches and the resistivity

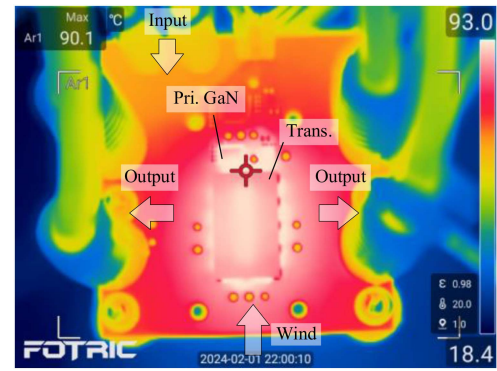


FIGURE 21. Thermal image at full load.

TABLE 4. Comparison of LLC DCXs with High Current and Ultra-Low Voltage Output

Solutions	V_{in} (V)	V_o (V)	I_o (A)	f_s (MHz)	Peak Eff. (%)	Full Load Eff. (%)	Power Density (W/in ³)	PCB Layers
This work	40	1	200	1	97	91.8	1394	6
[4]**	40	1	80	1	94.2	/	683	14
[5]	48	1	300	1	93.7	89	1010	18
[17]	40	1	130	1.2	92.7*	90.4*	443	/
[18]	36	0.75	150	1	94.1	86.8	580	4

*With driving loss.

**Only LLC DCX.

of the copper trace increase with temperature. Improved cooling can lower their temperature and reduce I^2R losses, thus improving efficiency at higher loads.

Fig. 21 shows the thermal image when the converter works at full load for two minutes. There is no heat sink. Air cooling is used, and the average wind speed is about 2.5 m/s. The hottest spot temperature is 90.1 °C near the output capacitors where the wind is blocked by the magnetic core. It can be predicted that the temperature rise of the devices and transformer will be reduced by further enhancing the cooling capability, such as adding a heatsink or using water cooling. The output current could be increased further.

Table 4 compares the performance of power converters with high conversion ratio transformers. The power density of all solutions includes driver circuitry. This work achieves the highest efficiency and power density, surpassing the other transformer-based solutions. Meanwhile, the number of PCB copper layers is only 6 in this work, which is beneficial for reducing costs and design complexity. The peak efficiency is 2.8% higher and the power density is almost doubled when compared with the matrix transformer solution [4]. It should be noted that the efficiency and power density of the solution [4] only consider the LLC DCX, and the buck and control circuit are removed for a fair comparison.

Fig. 22 shows the comparison of the efficiency curves. Since the load currents in the references are different, they are

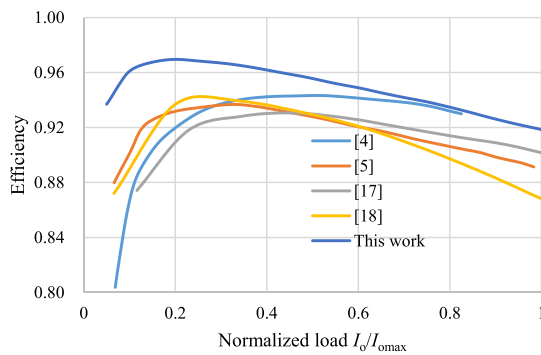


FIGURE 22. Efficiency comparison.

all normalized according to the maximum load current. This work has achieved high efficiency across the entire load range.

V. CONCLUSION

An integrated planar transformer is proposed for high-voltage-ratio LLC DCX with high current and ultra-low voltage output. The transformer has a secondary winding with a 1/10 turn, where the winding length is determined by the SRs and output capacitors. The constraints of the transformer structure are analyzed and a flexible compact structure is proposed. The transformer is implemented on a 6-layer PCB board, which can reduce cost and design complexity compared to other solutions with more PCB layers. Based on the proposed transformer, a 40 V–1 V/200 A, 1 MHz LLC DCX is built. Experimental results show that the LLC DCX achieves 97% peak efficiency, 91.8% full load efficiency, and 1394 W/in³ power density, surpassing the state-of-the-art transformer-based solutions. The proposed method is suitable for powering XPUs with ultra-low voltage and high current situations.

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